

Can high hatching/germination fraction drive extinction?

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Abstract

Invasive species often exhibit high propagule production (is there a friendlier general term?) under many different environmental conditions. Plant invaders often germinate at high rates across a suite of different temperature regimes, while native species appear to require highly specific cues and, even then, germinate at low rates. While this reality may explain why invasive species are so successful, it leaves open a massive question of what the hell the native species are doing. Here, we argue that answering this question requires reframing how we think about germination, survival and species extirpation alongside reframing the relevant timescales of environmental variation that we consider. We outline how the strategy of specifically timed, forever-low germination may be favored under certain climatic conditions. While such conditions have been uncommon in the short snapshot of ecological research they are more common over the timescales of species range shifts. We show how experiments and new analyses could begin to test this theory, with implications for restoration management.

Keywords: propagule pressure, seedbanks, bet-hedging

1 Misc notes I decided belong here

To do ...

1. Can we think of non-plant examples? Maybe sardines? Some small r-selected mammal? A bug? Megan thinks sardines are relatively short lived (3-5 years, reproductive in a year) and then their success is highly dependent on ENSO (and maybe they don't recover)
2. Resting stage and *Daphnia* – they think on century-long scale

Add in **biblical recruitment!** ... in 2014 in Hawaii

2 Notes to not forget from 2019 draft

There is also an evolutionary angle to this: short lifespans and short seedbanks with these sorts of long-term cycles could lead to evolving quickly, during one climate phase, towards a damn stupid strategy when the next climate phase kicks in. But I don't want to go there, just to point it out.

Opportunity to coin new term (from one term I hate and one that I like): **invasion extinction debt**, the number of invaders that would go extinct if you wait a very long time.

Refs from Doug: (Davison *et al.*, 2010; Morris *et al.*, 2008; Tuljapurkar *et al.*, 2009)

Lizzie's relevant refs on seed size, seed number and longevity: (Moles & Westoby, 2006; Rees, 1996; Saatkamp *et al.*, 2009; Silvertown, 1981; Thompson, 1987; Venable & Brown, 1988)

3 Outline

1. Introduction

- (a) Invaders are a problem
- (b) Invaders germinate a lot and easily
- (c) Native species do not (which seems weird)
- (d) Overview of some ways to try understand this paradigm (that don't really work)
 - i. Invaders are just superspecies who are traveling around with people now and taking over the world
 - ii. Humans have modified the world such that invaders are now successful (not sure how to make this work with germination)
 - iii. Something on the conditions under which invasive species evolve ...
 - iv. Native species are bet-hedging ... though this doesn't really make sense given how well invaders are doing ... unless ...
 - v. Invaders are not doing as well as you think everywhere, they are just doing well for now
- (e) Here we ...
 - i. examine how common invader strategies, such as high germination and germination plasticity, make them successful and under what circumstances do those strategies fail

2. How germination, germination, germination plasticity and (seedbank) survival relate

- (a) Germination and how it happens (define plasticity here) ... better term for germination plasticity: they have a broad germination niche, they are environmental generalists (vs specialist) in regards to germination conditions
 - (b) Seedbanks and how they happen (survival of seeds, but also how much you germinate matters) ... persistence vs 'losing' seeds to germination
 - (c) Germination and seedbank survival are inversely related – and this may be key for thinking about axes of potential strategies
 - (d) Compare what invaders vs natives do
3. When, if ever, does high germination not work out?
- (a) When high germination is followed by germinant/seedling failure (you use seeds but you don't get seeds back!) and you don't have another long-lived lifestage... sometimes
 - (b) We use a 'toy model' and show some results?
4. Section on timescales in invasion ecology (whole little section) ...
- (a) How fast does your seedbank die?
 - (b) What is the scale of your climate variation relative to that?
 - (c) (Maybe:) Does one favor faster evolution? Invader strategy will 'select hard on that.'
 - (d) (Maybe:) Compare long-term climate of where invader comes from vs non-native range climate.... no megadroughts in European Mediterranean?
5. What this all means for new studies and restoration ...
- (a) New studies should do (what)!?
 - i. Re think this: Maybe stop assuming there are ideal conditions that drive high germination in all species and assume some are actually designed to promote seedbank survival through variability of ideal conditions (I think maybe there is stuff out there on this?) ... a different type of bet-hedging (would this change how we model the seeds? Or would you end up with just a uniform distribution and 5% of seeds go every year?)
 - ii. Re-think about how you THINK about seedbank survival – there two ways out, die or germinate. Think more on germinate. (the relationship between the seedbank survival parameter and the idea of seedbank survival in models, then do new models!)
 - iii. Better measures of the relationship between climate that drives germination and leads to seed success vs failure (to figure out when they co-occur)
 - (b) Restoration implications
 - i. restoration through propagule pressure is already big: plow on native seeds and hope for the best
 - ii. our ideas here suggest this could be done better...

- iii. implications to target restoration with climatic conditions (though I bet CalHighway already does this)
- iv. Mow away the invaders!

- Davison, R., Jacquemyn, H., Adriaens, D., Honnay, O., de Kroon, H. & Tuljapurkar, S. (2010) Demographic effects of extreme weather events on a short-lived calcareous grassland species: stochastic life table response experiments. *Journal of Ecology* **98**, 255–267.
- Moles, A.T. & Westoby, M. (2006) Seed size and plant strategy across the whole life cycle. *Oikos* **113**, 91–105.
- Morris, W.F., Pfister, C.A., Tuljapurkar, S., Haridas, C.V., Boggs, C.L., Boyce, M.S., Bruna, E.M., Church, D.R., Coulson, T., Doak, D.F., Forsyth, S., Gaillard, J.M., Horvitz, C.C., Kalisz, S., Kendall, B.E., Knight, T.M., Lee, C.T. & Menges, E.S. (2008) Longevity can buffer plant and animal populations against changing climatic variability. *Ecology* **89**, 19–25.
- Rees, M. (1996) Evolutionary ecology of seed dormancy and seed size. *Philosophical Transactions of the Royal Society of London Series B-Biological Sciences* **351**, 1299–1308.
- Saatkamp, A., Affre, L., Dutoit, T. & Poschlod, P. (2009) The seed bank longevity index revisited: limited reliability evident from a burial experiment and database analyses. *Annals of Botany* **104**, 715–724.
- Silvertown, J.W. (1981) Seed size, life-span, and germination date as co-adapted features of plant life-history. *American Naturalist* **118**, 860–864.
- Thompson, K. (1987) Seeds and seed banks. *New Phytologist* **106**, 23–34.
- Tuljapurkar, S., Gaillard, J.M. & Coulson, T. (2009) From stochastic environments to life histories and back. *Philosophical Transactions of the Royal Society B-Biological Sciences* **364**, 1499–1509.
- Venable, D.L. & Brown, J.S. (1988) The selective interactions of dispersal, dormancy, and seed size as adaptations for reducing risk in variable environments. *American Naturalist* **131**, 360–384.

4 Figures

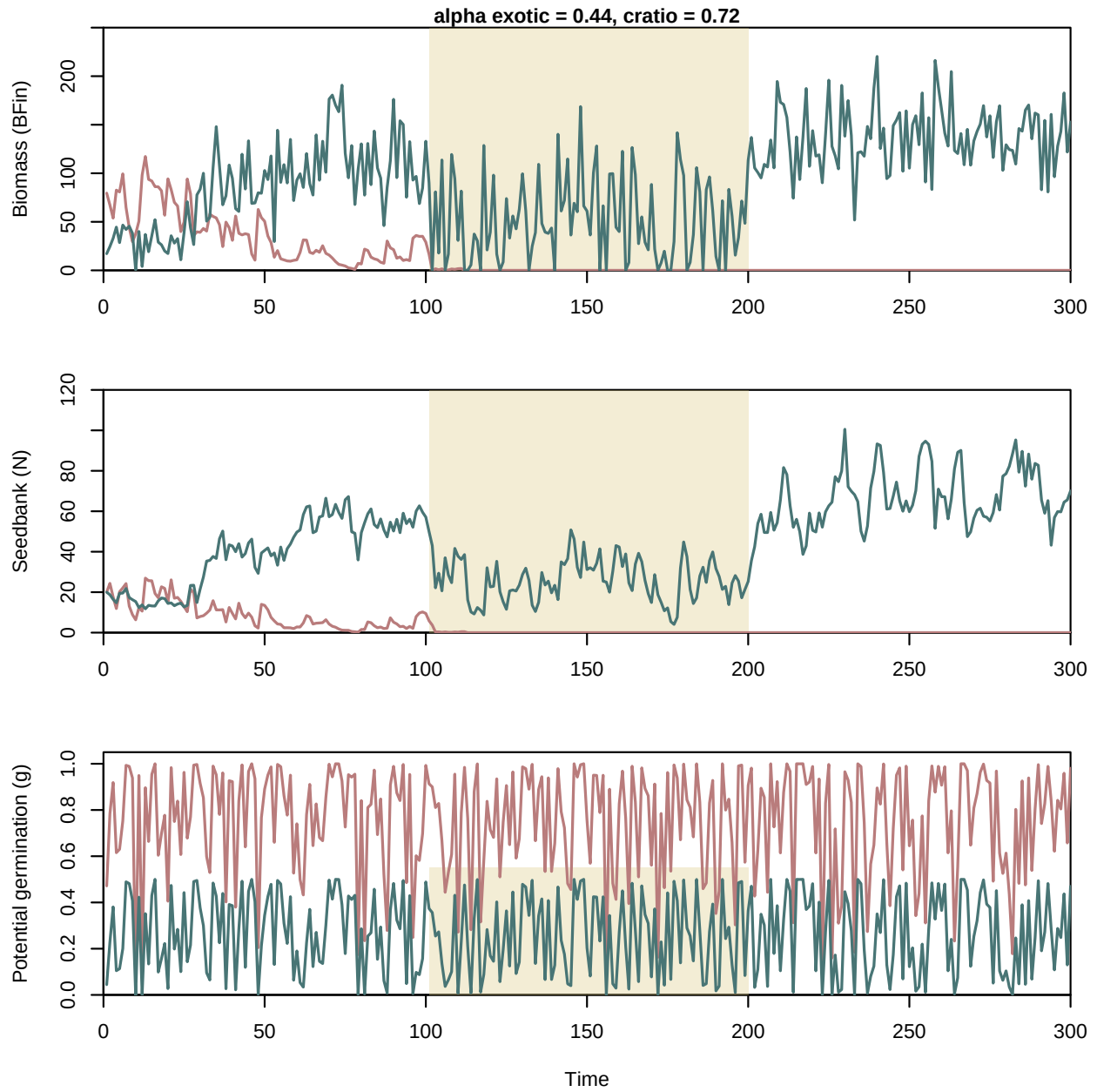


Figure 1: Burning out the seedbank!